

Program – 08: Create kubernetes Pods using YAML Descriptors

Structure of the program:

nodejs-configmap.yaml
nodejs-pod.yaml server.js

Steps:

1. Create a [server.js](#) file

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ nano server.js
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ nano node-config.yaml
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ nano node-pod.yaml
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ cat server.js
const http = require('http');
const port = 3000;

const server = http.createServer((req, res) => {
  res.send("Hello from node js running inside kubernetes.");
})

server.listen(port, () => {
  console.log("Server running at localhost:${port}");
})
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ cat node-config.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: node-app-config
data:
  server.js: |
    const http = require('http');
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.send("Hello from node js running inside kubernetes.");
    })

    server.listen(port, () => {
      console.log("Server running at localhost:${port}");
    })
```

2. Build the config map file

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ cat node-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-node
          mountPath: /usr/src/app/
      ports:
        - containerPort: 3000
  volumes:
    - name: app-node
      configMap:
        name: node-app-config
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ █
```

3. Build the pod yaml file

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ cat node-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-node
          mountPath: /usr/src/app/
      ports:
        - containerPort: 3000
  volumes:
    - name: app-node
      configMap:
        name: node-app-config
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ █
```

4. Apply the yaml files

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ kubectl apply -f node-config.yaml
configmap/node-app-config unchanged
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ kubectl apply -f node-pod.yaml
pod/program-8 created
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$
```

5. Check the status

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ kubectl apply -f node-config.yaml
configmap/node-app-config unchanged
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ kubectl apply -f node-pod.yaml
pod/program-8 created
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-2srn4   1/1     Running             1 (3m48s ago)  9m9s
nginx-deployment-6f9664446b-9k64x   1/1     Running             1 (3m48s ago)  9m9s
program-8                            0/1     ContainerCreating   0           36s
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$
```

6. Access the web page

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
labeled-pod   1/1     Running   1 (5h16m ago)  8h
my-pod        1/1     Running   1 (5h16m ago)  8h
program-8     1/1     Running   2 (37m ago)    10d
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
```

7. Check the output in:

<http://localhost:3000>

